AI SYNERGY HACKATHON 2026 / ROUND 1

Basic SRS / Implementation Plan

CivicTrace | Provisional Round 1 specification

SELECTED DIRECTION (PROVISIONAL)
CivicTrace — complaint ownership and traceable case history

Important: this is a working proposal, not proof that the hypothesised failure exists in any named municipality. The competition case's 23% reassignment figure is simulated case material, not an external real-world statistic.

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1. Purpose and problem boundary

CivicTrace is a proposed lightweight workspace for examining whether a municipal complaint's current accountable office and handling history can be understood across intake and departmental handoffs. The problem remains a hypothesis until a target workflow or records are examined. The prototype demonstrates traceability and human-supervised AI assistance; it does not claim to replace an official municipal platform.

Provisional problem statement

When complaints arrive through multiple channels or move between departments, the workflow may not consistently make the accountable office, assignment history, status changes and transfer reasons easy to trace. This may make it difficult for residents and authorised reviewers to determine who is responsible for the next action and how the complaint was handled.

2. Users and needs

User / actor	Need in Version 1
Resident / complainant	See the case ID, current stated owner, last recorded action, current status and correction/appeal route in a clear format.
Intake / grievance staff	Create a case and review structured-field suggestions without losing the original wording or media reference.
Department handler	Review the complaint, see prior actions, accept or question assignment, record a transfer reason and update status.
Coordinator / authorised reviewer	Reconstruct the timeline and see recorded reasons, corrections and human overrides.
Prototype evaluator	Run synthetic scenarios and score case-history reconstruction and safety-review behaviour.

3. Inputs and outputs

Inputs	Outputs
Complaint ID; original complaint text; chosen intake channel; language if known; optional synthetic attachment description; category/urgency suggestion; department assignment events; timestamps; action/status updates; transfer reason; correction/appeal event.	Case timeline; current accountable office or explicit "unconfirmed"; structured summary; suggested category with uncertainty; transfer-history view; human-review flag; status explanation based only on recorded events; audit/change history.

4. Workflow

- 1. Staff member or resident creates a case; the system generates a persistent ID and stores the original submission as entered.
- 2. Optional AI assistance extracts a short summary and suggests a category/department. The UI labels these as suggestions, not facts.
- 3. A human reviews, corrects or rejects the suggestions and confirms the accountable office. If ownership is unclear, the case remains "Needs human assignment".
- 4. Each handoff creates an event with from/to office, timestamp, reason and accepting/reviewing actor. The old event is not overwritten.

- 5. Status changes require a recorded action or an explicit reason. The interface shows the last recorded action, not a generated claim of progress.
- 6. Corrections, possible duplicate links, appeals and human overrides are appended to the history. The original case and separate reports remain preserved.
- 7. An authorised reviewer opens the timeline and attempts to reconstruct what happened.

5. Functional requirements

ID	Requirement	Priority / verification
FR-01	Create a complaint record with a persistent unique ID and preserve the original text.	Must; verify on 30 synthetic cases.
FR-02	Record current owner, assignment history, event timestamps and transfer reasons.	Must; verify timeline survives several handoffs.
FR-03	Allow “owner unconfirmed / needs human assignment” when available information is insufficient.	Must; test incomplete/ambiguous complaint.
FR-04	AI may suggest structured fields, a category and a short summary; the original submission remains visible.	Should; compare suggestion to labelled test cases.
FR-05	A possible duplicate is only suggested; a human must confirm linking. Linking must not delete either report.	Must; test two distinct residents reporting the same incident.
FR-06	Record status change, action note, correction, appeal and human override without erasing previous values.	Must; verify history can be reconstructed.
FR-07	Provide a reviewer view and a simple resident-facing status view that only uses recorded events.	Must; usability walkthrough.
FR-08	Provide a safety-review flag for seeded cases that meet the pre-defined test rule. AI cannot reject or close complaints.	Must; test all seeded safety scenarios.

6. AI and human responsibilities

AI may assist with	Humans remain responsible for
Drafting a concise summary; suggesting category/department; flagging a possible duplicate; raising a possible safety indicator; highlighting missing fields.	Confirming ownership and severity; deciding whether records describe the same incident; accepting/rejecting suggestions; transferring; prioritising; closing; responding to appeals; correcting factual errors.

Design rule: no AI-only rejection, closure, irreversible merge or final safety decision. Model output is untrusted until confirmed. Keep the original complaint beside any generated summary or translation.

7. Missing, conflicting or unreliable information

Condition	Required behaviour
Missing owner / conflicting departmental claims	Show “Needs human assignment”; preserve both claims and route to a named coordinator/reviewer if the workflow defines one. Do not invent an owner.
Low-confidence or contradictory AI suggestion	Display a clear “suggestion—verify” label or withhold the suggestion; require human review.
Possible duplicate but uncertain match	Do not merge automatically. Keep the cases separate and offer a review link.
Missing timestamp or reason	Show “not recorded”; do not fabricate. Permit an authorised person to add a correction with actor, date and reason.
Safety indicator or uncertain language interpretation	Flag for human review; preserve original language/input; do not downgrade or close solely due to model output.
AI service unavailable	Fall back to manual entry and assignment. Case history and status functions should still work.
Personal or sensitive data in test material	Use synthetic data by default. Use real records only with permission, minimisation and anonymisation.

8. Non-functional requirements and constraints

- Privacy: prototype uses synthetic examples; no public upload of citizen records or credentials.
- Explainability: show suggestions, uncertainty where available, human confirmation and recorded change reason.
- Audit trail: append events; avoid silently overwriting the original complaint or previous owner.
- Accessibility: readable labels, keyboard-usable core workflow and plain-language resident status.
- Feasibility: one small web demo with local sample data; avoid live municipal integration in Version 1.
- Reliability: core case-history workflow must remain usable when an optional AI API is unavailable.

9. Acceptance criteria

ID	Acceptance test	Pass condition
AT-01	Create 30 synthetic cases, including multi-department examples.	30/30 retain a stable ID, original complaint and reconstructable assignment timeline; current owner is identified or explicitly unconfirmed.
AT-02	Two reviewers independently reconstruct 20 synthetic histories.	At least 90% of reviewer-case assessments correctly identify current owner/status, transfer sequence and latest recorded action; disagreements are logged.
AT-03	Run seeded safety cases, ambiguous ownership, uncertain duplicates, missing fields and AI-outage scenarios.	No AI-only rejection/closure or irreversible merge; each required safety case is surfaced for human review; uncertainty is visible; manual fallback works.

10. Metrics and validation plan

Type	Measure	Interpretation
Baseline	Case-history reconstruction rate on existing records/workflow, if access is granted.	Share of cases where reviewers can identify owner, assignment sequence and latest meaningful action. If no real records are available, call it a synthetic/mock baseline.
Outcome	Change in correct reconstruction rate on the same cases after using CivicTrace; also record median review time.	Measures prototype traceability, not real-world resolution speed.
Harm / exclusion	Missed safety-review rate, overall and by English/Hindi/mixed-language test input.	A miss is a seeded safety scenario requiring human review that is not flagged. Small synthetic tests do not establish real-world safety.

Minimum validation before locking the statement: inspect one target municipality's public workflow; if permitted, request 30–50 anonymised case histories. Have two reviewers reconstruct them independently. If records are inaccessible, conduct a few consent-based workflow walkthroughs and use synthetic cases, explicitly labelling the result as feasibility testing rather than prevalence evidence.

11. Version 1 exclusions

- No integration with live municipal portals, phone systems, social media accounts or official databases.
- No automatic complaint rejection, closure, final priority decision or irreversible duplicate merging.
- No claim that CivicTrace reduces actual municipal resolution time or represents all Indian municipalities.
- No production-grade authentication, deployment, SLA, security certification or storage of real personal data.
- No autonomous interpretation of photos/voice notes for safety decisions; any such capability is outside scope unless specifically tested.

12. Implementation plan (small working slice)

Stage	Build / verify	Suggested owner
1. Data & test design	Define a simple case/event schema and 30 synthetic scenarios including transfers, missing fields, language variation, distinct duplicate-like reports and safety flags.	Member 1 + Member 3
2. Basic interface	Create case form, case list, case timeline, owner/status view and correction event entry.	Member 2
3. AI assistance	Add optional category/summary suggestion behind a clear human-confirmation step; provide manual fallback.	Member 3 + Member 2
4. Evaluation	Run acceptance tests; record all failures and reviewer disagreements; compare baseline and prototype on same cases.	All members
5. Demo packaging	Use synthetic data; show one normal case, one transfer, one ambiguous case, and one safety-review case.	All members

13. Proposed architecture

Web UI → application/workflow layer → case and event store. Optional AI assistance receives only the minimum complaint text needed to produce a suggestion and returns structured fields plus a confidence/uncertainty indication when supported. The workflow layer requires human confirmation before consequential changes. All changes append to the event history. The AI service is optional; manual workflow remains available.

14. Evidence and references

Case source: AI Synergy Hackathon 2026, Case 07 — "One Complaint, Five Departments" scenario supplied by the team. The 23% reassignment statistic is simulated competition material.

Aggarwal, V.; Krithika, P. R.; Garg, D. (2010). "Strengthening municipal governance for effective consumer grievance redressal." International Journal of Regulation and Governance, 10(2), 63–75. https://doi.org/10.3233/IJR-120095

Relevance: scholarly discussion of IT-enabled grievance systems in a few Indian cities; authors argue technology must be accompanied by accountability, transparency and capacity. Limitation: old, limited city sample, does not prove the specific case-record gap in our chosen target.

Department of Administrative Reforms and Public Grievances (DARPG), Comprehensive Guidelines for Handling the Public Grievances, Office Memorandum S-15/27/2024-PG Division.
https://www.darpg.gov.in/sites/default/files/Comprehensive_guidelines_for_handling_the_Public_Grievances.pdf

Relevant principles include transfer to the right authority rather than closing solely because it pertains elsewhere, detailed action-taken reporting, valid reasons for closure and appeal handling. Limitation: CPGRAMS/central-government guidance, not proof of a municipal system's actual implementation.

Comptroller and Auditor General of India (CAG), Audit Report on Panchayati Raj Institutions and Urban Local Bodies, Uttar Pradesh, year ended 31 March 2015, Report No. 2 of 2015.
https://cag.gov.in/uploads/download_audit_report/2015/Report_No2_of_2015_-_Panchayati_Raj_Institutions_and_Urban_Local_Bodies_Government_of_Uttar_Pradesh.pdf

Section 5.15 reports grievance-mechanism and complaint-register/monitoring deficiencies in specific test-checked local bodies. Limitation: historical, jurisdiction-specific evidence; not evidence of current nationwide prevalence or AI-related failure.

Srivatsa, V. et al. (2026). "Reimagining Digital Grievance Redressal Systems for Better Participatory Governance in India." DIS '26 Companion, ACM. https://doi.org/10.1145/3802974.3810897

Extended abstract describing a survey of 157 urban residents and interviews; discusses institutional silence and siloed teams. Limitation: extended abstract and study sample; does not establish that CivicTrace's specific failure exists in a target municipality.

15. Declarations to complete before upload

- Prototype status: Working software prototype not yet built. The current submission contains a problem brief, proposed architecture, presentation and implementation plan.
- Data: [Synthetic data only / other data with permission and anonymisation — state actual use].
- Reused assets: AI-assisted drafting and presentation preparation were used. No live municipal dataset or live municipal integration is included. The team should add any specific templates, software libraries, APIs, models or code actually used before submission.
- Team contributions (confirm before submission): Jhnavi Bhadkariya — team coordination and problem framing; Lavanya Bajpai — investigation and requirements; Aaryanvi Sharma — workflow, testing and technical planning. Edit this allocation if it does not match the work each member actually performed.
- AI interaction evidence: preserve the genuine chat exports/links and attach a separate decision index; do not rewrite original logs.